

Report

1. Regulatory Framework: The OFAC Sanctions List

The OFAC Sanctions List is a collection of publicly maintained databases administered by the Office of Foreign Assets Control, a financial intelligence and enforcement agency within the U.S. Treasury Department. Its primary function is to enforce U.S. economic and trade sanctions in alignment with national security and foreign policy goals by cutting targeted actors off from the U.S. financial system.

While commonly referred to as a singular list, it actually comprises several databases with varying levels of restriction. The targets on these lists generally fall into two main categories:

- **Comprehensively Sanctioned Jurisdictions:** Broad embargoes applied to entire countries or regions (such as Cuba, Iran, North Korea, Syria, Russia, and specific regions of Ukraine).
- **Targeted Individuals and Entities:** Specific people, companies, or transnational networks located anywhere in the world who are flagged for illicit activities, including narcotics trafficking, terrorism, weapons proliferation, and human rights abuses, or businesses acting as fronts for sanctioned regimes.

The regulatory framework is primarily enforced through two main databases:

- **The Specially Designated Nationals (SDN) List:** This is the strictest and most comprehensive database. It catalogs individuals, groups, and entities whose assets are immediately blocked (frozen) within U.S. jurisdiction. U.S. persons and businesses are strictly prohibited from conducting any business with them.
- **The Consolidated Sanctions List (Non-SDN):** This collection imposes highly targeted restrictions rather than blanket asset freezes. For example, the Sectoral Sanctions Identifications (SSI) List targets specific sectors of a foreign economy (such as finance or energy) without enacting a complete national embargo.

Compliance with OFAC regulations is mandatory for U.S. businesses and foreign entities dealing in U.S. dollars. Processing a transaction for a sanctioned individual—even accidentally—can result in severe civil and criminal penalties, often reaching hundreds of millions of dollars. This necessitates the integration of robust, automated OFAC screening protocols directly into modern transaction pipelines.

2. Categorization Criteria for Suspicious Transactions

To categorize ledger entries as suspicious or non-suspicious, the anomaly detection logic evaluates transactions against three primary risk vectors:

- **OFAC List Cross-Referencing (Geopolitical Risk):** The pipeline performs strict categorical checks on the counterparties and routing jurisdictions involved in the transaction. Any entity or routing country matching the OFAC SDN list or flagged as a sanctioned proxy triggers an immediate, maximum-confidence anomaly classification (a hard halt).
- **The Price Delta Feature (Trade-Based Arbitrage):** This continuous variable isolates potential over-invoicing or under-invoicing, a common typology in Trade-Based Money

Laundrying (TBML). The system calculates the spread between the executed unit price and the highly regulated global spot price for the asset:

$$\text{Price Delta} = \frac{|\text{Unit_Price_USD} - \text{Market_Spot_Price}|}{\text{Market_Spot_Price}}$$

Significant percentage deviations indicate potentially hidden fees, off-book transactions, or fabricated ledger entries.

- **Smurfing and Structuring (Volume Anomalies):** This statistical check flags high-frequency, low-volume behavior. The model identifies sequences where a single entity executes multiple transactions within a tight time window, intentionally keeping the total value of each transaction just below standard regulatory reporting thresholds to evade detection.

3. Mathematical Modeling: The Least Squares Method

The Least Squares method is utilized to fit a line (or hyperplane) to a set of data points by minimizing the total error between the observed data and the model's predictions. Specifically, it minimizes the **Sum of Squared Errors (SSE)**, placing a heavier penalty on significant outliers to ensure a robust fit.

For an n -dimensional input, the relationship is defined in matrix form as $y = X\beta + \epsilon$, where X is the design matrix, y is the target vector, β represents the weights, and ϵ is the residual error vector.

The loss function $L(\beta)$ is the sum of the squared residuals:

$$L(\beta) = \epsilon^T \epsilon = (y - X\beta)^T (y - X\beta)$$

Expanding this equation yields:

$$L(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta$$

To find the minimum loss, we take the gradient of $L(\beta)$ with respect to β and set it to the zero vector:

$$\frac{\partial L}{\partial \beta} = -2X^T y + 2X^T X \beta = 0$$

Rearranging this provides the Normal Equations:

$$X^T X \beta = X^T y$$

Multiplying both sides by the inverse of $(X^T X)$ yields the final, closed-form solution for the optimal weights:

$$\beta = (X^T X)^{-1} X^T y$$